

is incident over India's land area with most parts receiving 4-7 kWh per sqm per day. Solar photovoltaic power can effectively be harnessed providing huge scalability in India. National Institute of Solar Energy (NISE) has assessed the **country's solar potential of about 748 GW assuming 3% of the waste land area to be covered by Solar PV modules.** (MNRE website).

SECI unveils India's largest solar-battery project, pioneering renewable energy innovation in Chhattisgarh, 100 MW Solar PV project with 40 MW/120 MWh BESS. The project has deployed bifacial modules, which generate more electricity than monofacial modules, hence setting a new standard for large-scale renewable energy projects. In December 2023, **SECI commissioned the first on-grid solar project, with state-of-the-art Battery Energy Storage System (BESS), 1.7 MW with 1.4 MWh BESS in Lakshadweep islands.**

Both these projects use battery storage to supply electricity during non-solar hours. In addition SECI is also bringing a bigger tender of 1200 MW for solar plant with battery storage for 2 hours.

SECI is the implementing agency of **the Ultra Mega RE project in Ladakh with 13 GW with BESS in Leh District.** Currently various assessment studies are being carried out in this regard.

What have been the challenges and how have you overcome them while going in for expansion and the availability of infrastructure facilities and production capabilities to meet the demand for installation of solar energy panels?

India has been substantially depending on import of solar modules for installation of these plants. The very large capacities, way more than their domestic requirement in some foreign countries had created a situation where solar modules were very cheaply available in the international market and were hindering the growth of manufacturing industry. The country did not want to get into another kind of import dependence shifting from oil / gas dependent to solar module dependence. Hence, the Government started a Production Linked Incentive Scheme for National Programme on High Efficiency Solar PV Modules with an outlay of Rs. 24,000 crore. There are other challenges pertaining to Transmission Capacities, Land Laws & Regulation of the States and Regulatory issues which hinder the growth of the Renewable Energy sector. MNRE and SECI have been pro-active in sorting out the legal & regulatory issues, discussing with State Government to streamline and simplify the land related issues and making the provisions of Green Energy Corridors for easing out the transmission bottlenecks.

How is SECI promoting "Make in India" initiative by supporting private players and

start-ups in the solar energy sector, the incentives given to them and their energy contribution to the solar industry so far?

With the aim of **demand creation for domestic modules**, Government introduced the **scheme for setting up of 12 GW Grid- Connected Solar PV Power Projects by CPSUs** with domestic cells and modules. SECI is the implementing agency for **Phase II Tranche I and II (2026 MW of solar capacity)** of the scheme. VGF support is provided under this scheme. Under this tender **1655.50 MW has been already commissioned as on 30.04.2024.**

Further, to promote setting up of solar module/cell/wafer manufacturing facilities in the country **SECI has awarded 3 GW of solar manufacturing capacity under its manufacturing linked solar tender.** Under this tender **2 GW of manufacturing capacity has already been commissioned.**

SECI is the implementing agency for the **Production Linked Incentive (PLI) Scheme (Tranche-II)** Scheme for High Efficiency Solar PV Modules with an **outlay of Rs. 19,500 crore.** Under this tender, SECI has awarded **39600 MW of manufacturing capacities.** This Scheme has provision for PLI to the selected solar PV module manufacturers for five years post commissioning, on manufacture and sale of High Efficiency Solar PV modules.

India has diverse regional and climatic zones which makes for customized solar panels for installation. How is SECI addressing these issues be it survey, research for improving quality of the solar panel and battery, and the scientific know-how needed for installation of the solar panels?

SECI has introduced innovative models such as, solar-wind hybrid tenders, round-the-clock tenders, assured peak power supply tenders, standalone ESS, FDRE etc. :

➤ **Solar-wind hybrid tenders make use of the complementary nature of the solar and wind resources. Further it makes efficient use of the transmission infrastructure.**

➤ **Assured peak power supply tenders:** enable to address the challenges associated with grid stability and peak demand management

➤ **Round-the clock (RTC) power tender:** This tender combine power from RE generating sources with energy storage solutions to provide a **consistent power supply throughout the day.** Under this SECI has awarded 400 MW of project capacity.

➤ **Standalone Energy Storage systems:** This initiative will provide DISCOMs/ Grid operator with **storage capabilities that can be utilized as needed.** In pursuit of this objective, SECI awarded the **first-ever tender of its kind in the nation,** aiming to establish 500 MW/1000

MWh standalone Battery Energy Storage Systems (BESS) in India.

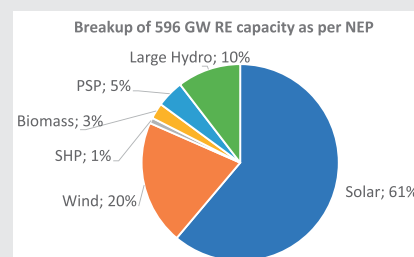
➤ **FDRE** ensures that energy buyers receive **power tailored to their needs**, these models signify a pivotal shift in the dynamics between energy buyers and suppliers. They must **adhere to pre-defined demand schedules** provided by buyers and ensure that **at least 90% of the specified demand** is met each month, with penalties imposed for any shortfalls in meeting these requirements.

In an amazing feat, India has surged ahead of Japan to become the third-largest solar power generator in the world in 2023, according to a report by global energy think tank Ember. From here, where is SECI and the green energy sector of India headed for?

Our Nation has set an ambitious target with '**Viksit Bharat @ 2047**', aiming to elevate India to the status of a developed nation by its centenary of independence. Central to this vision is the **Climate Agenda for 2047**, this agenda focuses on reducing emission intensity and boosting clean energy production.

India came out with **National Green Hydrogen Mission to make India the Global Hub for production**, usage and export of Green Hydrogen and its derivatives. The Mission will lead to significant decarbonisation of the economy, reduced dependence on fossil fuel imports, and enable India to assume technology and market leadership in Green Hydrogen.

National Electricity Plan (NEP) notified by Central Electricity Authority (CEA). **Likely installed capacity in India for the year 2031-32 is estimated to be around 900 GW (596 GW of Renewable based Capacity).**



SECI is expected to play a major role in achieving this target. SECI will be instrumental in the key structural interventions, including domestic manufacturing of RE technologies, promotion of energy storage, development of a standardized bidding framework, de-risking renewable energy investments, and ensuring secure payments—all crucial for meeting India's 2047 goals.